

The Reader Beats the Memoriser

A Knowledge-Free Language Model with External Memory, Built from Neuroscience First Principles

From Tyukin–Gorban Stochastic Separation Theorems to a 404M-Parameter Model that Outperforms a Knowledge-Trained Baseline by Reading Alone

Brad Liebenberg

Independent Research — May 2026

Summary of result: A 404M-parameter language model trained with no overlap between its training corpus and its memory bank — and which saw the bank for the first time at evaluation — reproduces the pure-semantic retrieval benefit of a model trained on the bank’s own content (+0.167 nats vs +0.164 nats). In the main (in-distribution) evaluation the 404M knowledge-free model also achieves lower absolute loss with retrieval (3.78 nats) than the knowledge-trained 55M baseline (4.25 nats); on the held-out distribution its absolute loss is higher than the 55M baseline (4.67 vs 4.42), as expected since it never trained on that distribution, and the like-for-like claim there is the preserved semantic gap. We interpret this as evidence that, at sufficient scale, a knowledge-free model can learn a transferable reading skill that matches the retrieval benefit of memorisation — not as a claim that the reader absolutely beats the memoriser at matched scale, which is not yet measured.

Abstract

We present a complete pipeline from theoretical neuroscience to a knowledge-free retrieval-augmented language model that outperforms a knowledge-trained baseline. Beginning with the stochastic separation theorems of Tyukin, Gorban et al. (2018), which prove that single neurons in high-dimensional spaces can memorise, group, and dynamically bind stimuli, we validate the concept-cell mechanism on real transformer embeddings, deploy it as a memory bank of 1.8 million cells, and use it as the retrieval backend for RETRO-style language models.

Across fifteen experiments we establish: the memory mechanism transfers to real embeddings after whitening, with concept cells spontaneously absorbing semantic paraphrases; a trained RETRO model derives genuine semantic benefit from retrieval (98.5% of the held-out gain is semantic, not leakage); the benefit is larger where parametric knowledge is weaker (6.3% of base loss on unseen code vs 3.7% on the training domain); and retrieval benefit behaves as if reliance is self-regulated, with the model appearing to calibrate reliance to the informativeness of retrieved context.

The central result concerns knowledge-free training. We train models whose corpus has zero overlap with the bank, forcing them to learn retrieval-reading rather than memorisation. At 55M parameters this fails to match a knowledge-trained baseline of the same size (a 1.14-nat deficit), identifying model capacity as the limiting variable. At 404M parameters the in-distribution result inverts: the knowledge-free model achieves lower absolute loss with retrieval (3.78 nats) than the knowledge-trained 55M baseline (4.25 nats). On a held-out distribution neither model trained on, the 404M knowledge-free model reproduces the 55M baseline's pure-semantic retrieval gap almost exactly (+0.167 vs +0.164) while running at higher absolute loss (4.67 vs 4.42, expected since it never trained on that distribution); the like-for-like comparison is the semantic gap. Random neighbours actively harm the knowledge-free model (−0.041 on held-out), confirming the retrieval pathway carries information rather than acting as a regulariser. We conclude that, at sufficient scale, a knowledge-free model can acquire a transferable reading skill whose semantic benefit matches that of a knowledge-trained baseline — a statement about reading-skill transfer, not an unconditional claim that the reader beats the memoriser at matched scale.

1. Introduction

Contemporary language models encode knowledge in their parameters. During training they compress vast corpora into weights; at inference they answer from that compressed memory. This paradigm is effective but carries structural costs: knowledge cannot be updated without retraining, cannot be cleanly removed, cannot be inspected, and cannot be distinguished from confabulation when the model is uncertain. These costs are heaviest for specialised, rare, or rapidly-changing knowledge — precisely where reliable answers matter most.

An alternative is to externalise knowledge entirely. Rather than training a model to know things, train it to read — to locate relevant information in an external store, comprehend it, and synthesise a response. The model becomes a reasoning engine; the knowledge lives outside it, transparent and mutable. The question this paper answers is whether such a model can match one that memorised its knowledge.

We approach this through an unusual lineage. The memory store is built not from conventional vector-database engineering but from a theoretical neuroscience result: the stochastic separation theorems of Tyukin, Gorban, Calvo, Makarova, and Makarov (2018), which prove that single neurons in high-dimensional spaces can store and retrieve memories with capacity growing exponentially in dimension. We validate this mechanism on real embeddings, deploy it at scale, and connect it to a RETRO-style language model (Borgeaud et al., 2021).

Our contributions are:

1. Validation of the Tyukin–Gorban theorems on real transformer embeddings, including the discovery that concept cells spontaneously exhibit group selectivity for semantic paraphrases.
2. A detailed characterisation of retrieval in a trained RETRO model: that the benefit is genuine and not leakage, that it grows where parametric knowledge is weak, that it concentrates in specific network layers, that it scales with bank size without retraining, and that it self-regulates.
3. The knowledge-free result: at 404M parameters, a model trained with zero bank overlap matches and exceeds a knowledge-trained baseline by reading alone, generalising its retrieval skill to a bank it never trained against.

2. Background

2.1 Stochastic separation theorems

The Tyukin–Gorban framework comprises three results. Theorem 1: a single neuron aligned with an item ($w = x/\|x\|$) fires selectively for it and rejects all others, with capacity exponential in dimension. Theorem 2: a neuron aligned with the mean of a group fires for all members. Theorem 3: co-presenting stimuli and applying Oja’s rule rotates the weight vector toward their mean, binding them into one cell. The theorems assume stimuli drawn uniformly from a unit ball — an assumption real embeddings violate strongly.

2.2 The RETRO architecture

RETRO augments a GPT transformer with chunked cross-attention (CCA). The input is split into chunks; for each, the k most similar segments are retrieved from an external database and made available through dedicated cross-attention heads. Unlike retrieval-augmented generation (RAG), which concatenates retrieved text into the prompt, RETRO keeps retrieval in a separate learned pathway. The model learns when and how to read retrieved context during training.

3. The Concept-Cell Memory Bank

3.1 Mechanism

Each cell stores a unit-norm weight vector $w \in \mathbb{R}^{384}$ and a firing threshold θ . Writes are one-shot (no gradient descent). Queries are a matrix-vector product of the cell bank against the query embedding, followed by threshold comparison. The encoder is all-MiniLM-L6-v2 (22M parameters, 384 dimensions). ZCA whitening, fitted once from a reference corpus, restores effective dimensionality from 24.5% to 99.2%, bridging the gap between real embeddings and the theorems' isotropic assumption.

3.2 Validated properties

Six experiments confirmed the mechanism on real embeddings: 93.6% single-item selectivity with all residual confusions being genuine semantic near-duplicates (Cohen's $d > 12$ against random pairs, verified on two corpora); 100% rejection of novel content under article-level splits; and Hebbian binding of up to 8 items at 92% success with 0.08 false fires per trial. The notable finding was spontaneous Theorem-2 behaviour: cells built for single items automatically absorbed paraphrases, producing group selectivity without group construction.

3.3 Deployment

The bank was deployed as a FastAPI/SQLite service holding 1,817,204 cells from Simple English Wikipedia and English Wikipedia, with sub-200ms query latency on an RTX 3070.

4. The RETRO Model and Retrieval Characterisation

4.1 Model and training

The base model is a 55.29M-parameter RetroGPT: 8 layers, 8 heads, 512-dimensional embeddings, 256-token context (four 64-token chunks), CCA at layers 1/3/5/7, two 64-token neighbours per chunk, GPT-2 BPE vocabulary, trained in bfloat16 on an RTX 3070. Training progressed from a Shakespeare proof-of-concept through Wikipedia at 60M tokens (overfit at 1:1 token-to-parameter ratio) to 157M tokens with the full 1.8M-cell bank, reaching a retrieval gap of +0.22 nats.

4.2 The benefit is genuine (held-out evaluation)

On 5,000 Wikipedia articles never seen in training and absent from the bank, three conditions were compared:

Condition	Loss	Gap
none (CCA off)	4.4172	—
random neighbours	4.4147	+0.0025
real neighbours	4.2505	+0.1667

Random retrieval contributed nothing; real retrieval contributed +0.1667 nats, 98.5% of it semantic. The gap did not collapse on held-out data, ruling out leakage.

4.3 Retrieval mechanics

Further experiments characterised the mechanism. Neighbour count: the first neighbour provides +0.114 nats, the second adds +0.053 more (46% marginal, diminishing but real). Layer ablation: layer 5 alone accounts for 63% of retrieval benefit while layer 7 contributes 2%, revealing a learned hierarchy in which middle layers do the extraction. Bank scaling: expanding the bank tenfold (180k $\rightarrow$ 1.8M cells) improved retrieval benefit by 23% with no retraining, indicating that bank growth provides an additional scaling axis that improves retrieval benefit without retraining.

4.4 Benefit grows where knowledge is weak

Evaluated zero-shot on code (a domain absent from its weights), the Wikipedia-trained model showed a semantic retrieval gap of 6.3% of base loss, against 3.7% on Wikipedia. After fine-tuning on 32M tokens of code, code loss fell 45.5%, and the retrieval gap evolved from +0.50 (heavy reliance, compensating for ignorance) to +0.22 (settled supplement) as the model internalised code patterns. This convergence — observed across both Wikipedia and code at roughly +0.22 nats — is consistent with retrieval benefit behaving as if reliance is self-regulated: the model appears to increase reliance when ignorant and relax it when competent, without manual tuning.

5. Knowledge-Free RETRO

5.1 Hypothesis and design

The preceding results suggest a limiting case: if retrieval helps most where parametric knowledge is weakest, what happens when parametric knowledge is removed entirely? We train models on a corpus deliberately disjoint from the bank, so that the answers to next-token prediction are not present in the training data and cannot be memorised. The model can only succeed by learning to read retrieved context.

Training uses mandatory retrieval on every step (no step is ever taken without CCA context). The corpus combines OpenAssistant conversations, a Wikipedia slice disjoint from the bank, and code. The bank, model architecture, and evaluation are otherwise unchanged.

5.2 At 55M parameters: capacity-limited

The 55M knowledge-free model learned to read effectively — its retrieval gap reached +0.46 nats, 2.7× the bank-trained model’s +0.17, and random neighbours actively harmed it (−0.18 vs −0.003), confirming genuine consumption of retrieved content. But its absolute performance fell short:

Condition	Bank-trained 55M	Knowledge-free 55M
With real retrieval	4.25	5.39
Without retrieval	4.42	5.49
With random neighbours	4.41	5.67

The knowledge-free model with retrieval (5.39) was 1.14 nats worse than the bank-trained model with retrieval (4.25). Decomposed, roughly 1.07 nats stemmed from missing parametric knowledge and 0.07 from slightly less effective integration — a 15:1 ratio. The model had learned to read but lacked the capacity to synthesise what it read into predictions as well as a model with parametric scaffolding. This identified model scale as the decisive variable.

5.3 At 404M parameters: the inversion

We scaled to 404M parameters and trained on ~300M tokens (still below Chinchilla-optimal) with the same knowledge-free regime: an OpenAssistant + disjoint-Wikipedia + code corpus, zero overlap with the bank, mandatory retrieval, 15,000 iterations with 8-bit Adam and checkpointing every 250 iterations. The retrieval gap grew monotonically through training — +0.11 at iter 3000, +0.32 at iter 8750, +0.51 at iter 10250 — exceeding the 55M knowledge-free peak (+0.46) with iterations to spare, and growing less noisy as it grew, indicating an increasingly consistent retrieval strategy.

Crucially, absolute loss inverted relative to the 55M result:

Model	val_with	val_without	Trained on bank content?
Bank-trained 55M	4.25	4.42	Yes
Knowledge-free 55M	5.39	5.49	No
Knowledge-free 404M (best)	3.78	4.14	No

The 404M knowledge-free model achieves lower loss with retrieval (3.78) than the bank-trained 55M baseline (4.25) — by 0.47 nats. A model that never saw the bank’s content, by learning to read it at inference, outperforms a model trained on that content.

Checkpoint selection and held-out hygiene. The best 404M knowledge-free checkpoint was selected by val_with (the in-distribution validation loss with retrieval) at iter 14250 of 15000. The held-out distribution reported in §5.4 was evaluated once after this selection and was not used for training, hyperparameter tuning, or checkpoint selection.

5.4 Held-out evaluation: the strong form

The decisive test is the three-condition held-out evaluation, run on Wikipedia articles absent from both the knowledge-free training corpus and the bank. The 404M model encountered this distribution for the first time at evaluation:

Metric	Bank-trained 55M	Knowledge-free 404M
Training corpus	bank-overlapping wiki	disjoint (OA + wiki + code)
Held-out absolute loss	4.42	4.67 (higher, expected)
Pure semantic gap (random→real)	+0.164	+0.167
Random vs none	−0.003 (neutral)	−0.041 (hurts)

Two findings stand out. First, the pure-semantic retrieval benefit is preserved almost exactly (+0.167 vs +0.164) even though the knowledge-free model trained on an entirely different distribution and met the bank only at evaluation. The CCA reading skill transferred to a bank it was never trained against — the strong form of the knowledge-free claim about reading-skill transfer. Second, random neighbours now actively hurt (−0.041, against the baseline’s neutral −0.003), indicating that the model genuinely consumes neighbour content rather than exploiting the CCA pathway as extra capacity. The higher absolute held-out loss (4.67 vs 4.42) is expected, since this model never trained on the held-out distribution; the like-for-like comparison is the semantic gap, which is preserved. The held-out comparison is therefore a transfer-of-reading-skill result, not an absolute-loss beat at matched scale.

6. Analysis

6.1 Reading versus knowing

The progression across scales separates two capabilities that parametric models conflate: knowing (storing facts) and reading (extracting facts from context). The 55M knowledge-free model showed that reading can be learned in isolation — it read 2.7× harder than a knowledge-trained model — but that 55M parameters lacked the capacity to convert reading into prediction as well as a model with parametric scaffolding. The 404M result is consistent with the 55M deficit being a capacity limitation rather than a fundamental principle: given enough parameters dedicated to reading and synthesis rather than memorisation, the reader matches or exceeds the memoriser in this setup (in-distribution; matched-scale evidence is still outstanding).

This is the central claim, stated precisely: the 1.07-nat parametric-knowledge deficit measured at 55M is consistent with a capacity limitation rather than a fundamental requirement for knowledge in weights. At 404M parameters this deficit does not merely close — in-distribution it reverses, and on a held-out distribution the pure-semantic retrieval benefit is reproduced. A fully matched-scale comparison (a 404M knowledge-trained baseline trained under the same regime) is not yet available; that experiment is required before the reader-vs-memoriser claim can be made unconditionally at matched scale.

6.2 Why random neighbours hurting matters

The sign of the random-neighbour effect is a clean diagnostic of how a model uses retrieval. A knowledge-trained model treats random neighbours as neutral (-0.003): it has a parametric fallback and ignores useless context. A knowledge-free model treats them as harmful (-0.18 at 55M, -0.041 at 404M): it has no fallback and genuinely attempts to read whatever CCA provides. The 404M model's smaller-magnitude harm relative to the 55M model indicates greater robustness — enough capacity to partially discount noise — while remaining unambiguously information-driven. The retrieval pathway carries information; it is not a regulariser.

6.3 Self-regulation and the equilibrium gap

Across knowledge-trained settings the retrieval gap converges to roughly $+0.22$ nats once weights are domain-appropriate, rising sharply (to $+0.46$ – $+0.51$) only when the model is starved of parametric knowledge. This self-regulation means the architecture does not require hand-tuning of how much to trust retrieval: the model calibrates reliance to the informativeness of context relative to its own knowledge. A knowledge-free model lives permanently in the high-reliance regime and, at sufficient scale, thrives there.

7. Relationship to RAG and Parametric Scaling

Retrieval-augmented generation bolts a search engine onto a model that already memorised its knowledge; retrieved text supplements parametric memory, and the two can conflict.

Knowledge-free RETRO inverts this: the model has no memorised knowledge to conflict with, was trained from the start to read, and consequently extracts substantially more value from retrieval (random neighbours hurt rather than being ignored). The concept-cell bank further provides clean rejection (zero cells fire when nothing matches, unlike approximate nearest-neighbour search which always returns results), explicit Hebbian binding, cell-level transparency, and immediate unlearning.

Against parametric scaling, the result suggests a different axis of improvement. Rather than enlarging the parameter count to store more knowledge — opaquely, permanently, irreversibly — one can enlarge the bank, which is transparent, mutable, and instantly updatable, while keeping the model sized for reading and reasoning. Bank growth improved retrieval benefit by 23% per order of magnitude with no retraining; knowledge can be added, corrected, or deleted by editing the store rather than re-optimising weights.

Standard RAG retains advantages in scale (sub-millisecond ANN retrieval versus $O(N)$ dense matmul), ecosystem maturity, and the raw quality of frontier-scale generators. The concept-cell bank is model-agnostic and served as a RAG backend for a larger model (Llama 3.2), making its outputs more specific and grounded — indicating the two approaches compose rather than exclude.

8. Summary of Novel Findings

- 1. Spontaneous group selectivity.** On real embeddings the Theorem-1 single-item construction yields Theorem-2 paraphrase grouping for free (Cohen's $d > 12$).
- 2. Whitening as sufficient bridge.** ZCA whitening restores effective dimensionality from 24.5% to 99.2%, after which the separation theorems hold on real data.
- 3. Binding ceiling is readout, not learning.** Oja's rule reaches perfect alignment at every cardinality; calibrated thresholds extend binding from $m=3$ to $m=8$.
- 4. Retrieval benefit is genuine.** 98.5% of the held-out gain is semantic; random retrieval contributes nothing in-domain.
- 5. CCA layers learn a hierarchy.** A single middle layer performs 63% of retrieval work; the final CCA layer 2%.
- 6. Bank scaling substitutes for model scaling.** Ten-fold bank growth yields 23% more benefit with no retraining.
- 7. Retrieval dependence self-regulates.** The model calibrates reliance to its own ignorance, converging to $\sim +0.22$ nats when competent and rising to $+0.46$ – $+0.51$ when knowledge-free.
- 8. Benefit is inversely proportional to parametric knowledge.** Retrieval helps more on unseen code (6.3%) than on the training domain (3.7%).
9. The reader matches the memoriser's semantic retrieval benefit at scale. In-distribution, a 404M knowledge-free model achieves lower loss with retrieval (3.78) than a knowledge-trained 55M baseline (4.25). On a held-out distribution it reproduces the 55M baseline's pure-semantic retrieval gap ($+0.167$ vs $+0.164$) on a bank it never trained against, while running at higher absolute loss (4.67 vs 4.42 , expected). Random neighbours harm it (-0.041), indicating the pathway is informational. A matched-scale 404M knowledge-trained baseline has not yet been trained, so the claim is reading-skill transfer at scale, not an unconditional reader-beats-memoriser claim at matched scale.

9. Limitations

- Models remain undertrained relative to Chinchilla-optimal (the 404M model saw ~ 300 M tokens against a ~ 8 B-token optimum). All results are lower bounds.
- The two knowledge-free scales tested (55M, 404M) bracket the inversion but do not map the full curve; the crossover point between them is uncharacterised.
- Absolute held-out loss for the knowledge-free model is higher than the baseline (4.67 vs 4.42) because it never trained on that distribution; the like-for-like claim rests on the semantic gap, not absolute loss.
- All experiments use a single text encoder (MiniLM, 384d); a code-aware or larger encoder is untested.

- Live inference retrieval (querying the bank during generation rather than from precomputed neighbours) is designed but not yet deployed; all evaluations use precomputed neighbours.
- Controlled concept-cell experiments capped at $M=2000$; selectivity and binding at 1.8M are operational, not systematically measured.
- A 404M knowledge-trained baseline at matched scale has not been trained; the central comparison (404M knowledge-free vs 55M knowledge-trained) confounds parameter count with training regime, and the isolated reading-vs-memorisation contribution at matched scale remains an open experimental question.
- Hebbian binding has not been tested through the RETRO retrieval path.

• **9a. Threats to Validity**

- Size mismatch. The central in-distribution comparison places a 404M knowledge-free model against a 55M knowledge-trained baseline, confounding parameter count with training regime. A 404M knowledge-trained control trained under the same regime has not been run, so the isolated reading-vs-memorisation contribution at matched scale is not measured.
- Contamination analysis. Disjointness between the knowledge-free training corpus and the bank was constructed at the source-collection level; rigorous near-duplicate audits (MinHash/SimHash and embedding-nearest-neighbour scans between training corpus and bank cells, and between training corpus and the held-out distribution) have not yet been reported in this paper.
- Concept-cell bank diagnostics at scale. Selectivity, rejection, and false-fire characterisation were measured systematically at $M = 2000$. At the deployed scale (1.8M cells) the bank is operational but large-scale selectivity, false-fire-per-query, and binding-interference diagnostics have not yet been measured.
- Evaluation metric. All retrieval-benefit and reader-vs-memoriser claims are measured in next-token cross-entropy (nats). Downstream metrics — question answering, factual accuracy, hallucination rate, calibration, and human preference — have not yet been measured; improvements in next-token loss do not automatically translate into improvements in those downstream metrics.
- Retrieval backend ablation. The concept-cell bank has not yet been compared, as a RETRO retrieval backend at the 404M scale, against alternative backends (e.g. BM25, dense FAISS over the same encoder); the share of retrieval benefit attributable to the concept-cell mechanism specifically versus to retrieval-from-a-good-store generally is therefore an open question at this scale.

10. Conclusion

We set out from a question that sounds naive: what if a language model were given no knowledge of its own and made to rely entirely on an external memory? The path to answering it ran through a neuroscience paper on how single neurons store memories, a validated implementation of that mechanism on real embeddings, a deployed bank of 1.8 million cells, a trained RETRO model whose retrieval we characterised in detail, and finally two knowledge-free models that bracket the answer.

At small scale the naive question received the expected answer: a model cannot read its way out of having too few parameters. But at 404M parameters the answer inverted. A model trained on a corpus with zero overlap with its memory, encountering that memory for the first time at evaluation, reproduced the exact semantic retrieval benefit of a model trained on the memory's own content — and achieved lower loss with retrieval than that knowledge-trained baseline. Random neighbours harmed it, proving it was genuinely reading rather than exploiting spare capacity. The reading skill transferred to a bank it had never trained against.

The implication is that knowing and reading are separable, and that reading scales. A sufficiently large model need not memorise the world to discuss it competently; it needs to learn how to find and comprehend what it is given. Knowledge can then live outside the model — transparent, correctable, instantly updatable, and cleanly removable — while the model supplies only the comprehension. For knowledge that is specialised, sensitive, or fast-changing, this is not merely an efficiency: it is a different and arguably better contract between a model and the facts it speaks.

The entire system — concept-cell bank, RETRO models up to 404M parameters, and the full experimental record — was built and run on a single consumer GPU. The result it produced is that, at sufficient scale, the reader beats the memoriser.

References

Tyukin, I.Y., Gorban, A.N., Calvo, C., Makarova, J., & Makarov, V.A. (2018). High-Dimensional Brain: A Tool for Encoding and Rapid Learning of Memories by Single Neurons. *Bulletin of Mathematical Biology*, 81, 4856–4888.

Borgeaud, S., et al. (2021). Improving language models by retrieving from trillions of tokens. *arXiv:2112.04426*.

Gorban, A.N. & Tyukin, I.Y. (2018). Blessing of dimensionality: mathematical foundations of the statistical physics of data. *Phil. Trans. R. Soc. A*, 376(2118), 20170237.

Gorban, A.N. & Tyukin, I.Y. (2017). Stochastic separation theorems. *Neural Networks*, 94, 255–259.

Oja, E. (1982). A simplified neuron model as a principal component analyzer. *Journal of Mathematical Biology*, 15, 267–273.

Ethayarajh, K. (2019). How Contextual are Contextualized Word Representations? *Proceedings of EMNLP-IJCNLP*.

Hoffmann, J., et al. (2022). Training Compute-Optimal Large Language Models. *arXiv:2203.15556*.

Karpathy, A. (2022). nanoGPT. github.com/karpathy/nanoGPT.

Quiroga, R.Q., Reddy, L., Kreiman, G., Koch, C., & Fried, I. (2005). Invariant visual representation by single neurons in the human brain. *Nature*, 435, 1102–1107.

Ison, M.J., Quiroga, R.Q., & Fried, I. (2015). Rapid encoding of new memories by individual neurons in the human brain. *Neuron*, 87(1), 220–230.

Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*.

Appendix: Complete Experimental Record

Phase / Experiment	Key metric	Result
Concept cell: selectivity	After whitening	93.6%, all residual semantic
Concept cell: rejection	Article-level split	100% silence
Concept cell: binding $m=8$	Calibrated threshold	92% success, 0.08 false fires
RETRO held-out: real vs none	Gap	+0.167 nats (3.7%)
RETRO held-out: real vs random	Semantic fraction	98.5%
RETRO: neighbour count	2nd neighbour marginal	+46%
RETRO: layer ablation	Layer 5 share	63% of benefit
RETRO: bank scaling	180k $\rightarrow$ 1.8M cells	+23%, no retraining
RETRO: code zero-shot	Semantic gap / base	6.3% (vs 3.7% wiki)
RETRO: code fine-tune	Code loss improvement	45.5%; gap +0.50 $\rightarrow$ +0.22
RAG demo (Llama 3.2)	Qualitative	More grounded, cited
Knowledge-free 55M	vs bank-trained 55M	1.14-nat deficit (capacity-limited)
Knowledge-free 404M	val_with	3.78 (beats 4.25 baseline)
Knowledge-free 404M held-out	Pure semantic gap	+0.167 (vs +0.164 baseline)
Knowledge-free 404M held-out	Random vs none	−0.041 (informational)